

AUTONOMOUS CROP-SCOUTING ROVER

AgriRover

A rugged, low-cost field rover that walks tomato rows for smallholder farms, captures **geotagged crop evidence**, runs **on-device AI** for weeds, disease and obstacles, and hands an agronomist a reviewable record — instead of asking a farmer to guess.

TEAM

Hitanshu Kapadiya	25B2228
Vivek Kumar Gupta	25B2269
Shreyash Wagh	25B2227

B.Tech Mechanical Engineering · IIT Bombay
Pilot cluster: Narayangaon / Junnar, Maharashtra

01 PROBLEM & APPROACH



Prototype rover on a close-row scouting pass. Tank drive, pan-tilt camera, GPS mast, rear utility deck.

- **Scouting does not scale by walking.** Pest and disease outbreaks in tomato are caught late because nobody inspects every row every week.
- **Spray decisions are made blind.** Without dated, located evidence, treatment turns into calendar spraying — cost, residue and pre-harvest-interval risk.
- **Nothing is written down.** No image trail, no treatment log, no way to check whether an intervention actually worked.

THE SERVICE LOOP WE AUTOMATE

01
Sentinel samples

02
Trap images

03
Triggered close-
row scan

04
Agronomist review

05
Treatment + PHI
record

06
Outcome
verification

02 SYSTEM ARCHITECTURE — DUAL CONTROLLER

REAL-TIME TIER

ESP32 DevKit V1

FreeRTOS · dual core

- 2× BTS7960 tank drive, LEDC PWM
- NPK/RS-485, DHT22, moisture, TDS
- GPS, battery and current sensing
- Sequential pump + actuator dosing
- MQTT telemetry over WiFi

HMAL-AUTH
UART LINK

PERCEPTION TIER

Raspberry Pi 5 + Hailo-8 AI HAT+

26 TOPS · Coral USB fallback

- Obstacle / weed / leaf-disease inference
- EKF pose fusion: GPS + encoders + IMU
- Geotagged evidence capture & plant DB
- Active-learning frame capture, model OTA
- Streamlit dashboard + Pathway streaming

MQTT / TLS
+ TELEGRAM

OPERATOR TIER

Phone & field ops

any handset, no app install

- Two-way Telegram: /stop /go /status /photo
- Live dashboard, map and scan reports
- Push alerts on detection or fault
- ISOXML export path (ISO 11783)
- systemd services, hardware watchdog

03 ENGINEERING THAT MAKES IT SURVIVABLE

Layered stop logic

- VL53L1X ToF proximity cut-off
- Obstacle inference veto on drive
- IMU tilt and rollover threshold
- Heartbeat watchdog on the link
- Motor-stall current detection

Field robustness

- Thermal guardian: CPU, pack, ambient
- Fuel gauge + current monitor budgeting
- Encoder + IMU odometry for row keeping
- Simulation rig — full stack, no hardware
- pytest suite over AI, nav, data, safety

Trust & security

- HMAC-authenticated, encrypted commands
- MQTT over TLS with broker auth
- ESP32 secure-boot path, hardened Pi image
- Immutable evidence and treatment records
- Published threat model in SECURITY.md

04 HONEST STATUS & VALIDATION GATES

Production-intent prototype

Firmware, Pi services, ROS 2 nodes, CAD, wiring, a costed BOM and shopping list, simulation and model artifacts are in the repository. Field evidence for crop accuracy, acres/day, chemical reduction and payback is deliberately **not yet claimed**. The first pilot keeps spraying manual and does not prescribe fertilizer from the low-cost NPK probe.

Software, firmware & simulation



built

Wiring, BOM & mechanical design



built

Supervised physical traversal



in progress

Local tomato dataset + per-class val.



gate

12-grower paired pilot, KVK review



gate